Pairwise Analysis of Correlation Between Genetic Differentiation and Divergence in House Sparrows

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1 Introduction

To examine the relationship between genetic differentiation and divergence for correlation, those metrics must first be understood. Differentiation, or *fixation index* (F_{ST}), is an estimation of population differentiation between two populations, normalized by each population's own differentiation. Divergence, or in the case of this paper, *per-site divergence* (d_{XY}) is the direct nucleotide divergence between populations. They are calculated in fundamentally different ways. F_{ST} is defined as the average frequency of an allele among different populations, balanced against the diversity of a population itself, while d_{XY} is defined simply as the proportion of sites that differ between populations. In other words, one takes into account a populations diversity when doing pairwise comparisons, while the other does not. To determine if these are correlated, we built a pipeline to answer this question. The pipeline consists of filtering, data conversion, parameter and statistics calculations, and plotting using a variety of bioinformatic tools.

2 Methods

A shell script was used to filter, index, and convert a VCF file. The filtering step was applied using hard filters in VCFtools at both the genotype and variant level in lieu of a truth set [1]. Only biallelic SNPs passed the filter. Any genotypes with Phred scores below 20 was set to missing, as well as any genotypes with sequencing depths outside the range of 5–70. Then, sites with missingness of 10% or more were excluded. After filtering, data were indexed using BCFTools [2]. Once indexed, they were converted first to an Intermediate Columnar Format (ICF) directory, then to a Zarr directory using vcf2zarr [3].

A Python script was used to read the Zarr data into memory, compute F_{ST} and d_{XY}, calculate correlation between the two metrics, and plot the results. The filtered Zarr directory was assigned to an Xarray Dataset object using sgkit [4, 5]. A new dimension was merged into the dataset as samples were divided into cohorts based on their populations. The dataset was then divided into two Xarray Datasets, one for each chromosome in the VCF. Windows of 300 variants were then merged into the dataset, and the variant dimension, an Xarray DataSet, was chunked to match the size of the windows, at 1,250 variants per chunk, allowing pairwise F_{ST} and d_{XY} to be calculated and stored in similarity matrices, one for each window of variants using sgkit. Finally, Spearman's ρ was calculated on groups of pairwise F_{ST} and d_{XY} sets for each chromosome using SciPy, and data were plotted using matplotlib[6, 7].

3 Results & Conclusions

Filtering cutoffs were motivated by close examination of the data as well as the method outlined by Dennis et al., who motivated the decision to include only biallelic SNPs [8]. sgkit was used to generate F_{ST} and d_{XY} because of its flexibility with assigning cohorts and windows, integration of Xarray's strengths like chunking, and because it is the successor to scikit-allel, a popular toolkit that is now only being maintained. Windowing by variant was chosen for its reliability in the number of sites per window, guaranteeing 1,250 VCF rows of data for each window on which F_{ST} and d_{XY} were computed, except for the final window of each chromosome. A window size of 1,250 variants was chosen because base pairs outnumbered sites in the original data by roughly 40:1, meaning that roughly the same number of windows would be obtained using 1,250-variant windows as there would be 50kb-position windows. Spearman's ρ was used to assess correlation because upon scrutinizing the $F_{ST} - d_{XY}$ plots, many groups appeared right-skewed along F_{ST}, thus violating the assumptions necessary for computing Pearson correlation coefficients.

There was more meaningful correlation between F_{ST} and d_{XY} for some populations compared to others (Figure 1). For instance, differentiation and divergence appeared highly correlated between both Lesina and 8N and Lesina and K cohorts on chromosome Z, with $\rho = 0.94$ and $\rho = 0.92$ respectively ($p = 3 \times 10^{-91}$, $p = 1 \times 10^{-80}$). Spearman's ρ was estimated with high certainty for all cohort comparisons and chromosomes ($p < 0.05$), but for the remainder of the cohort pairs, correlation was less clear.

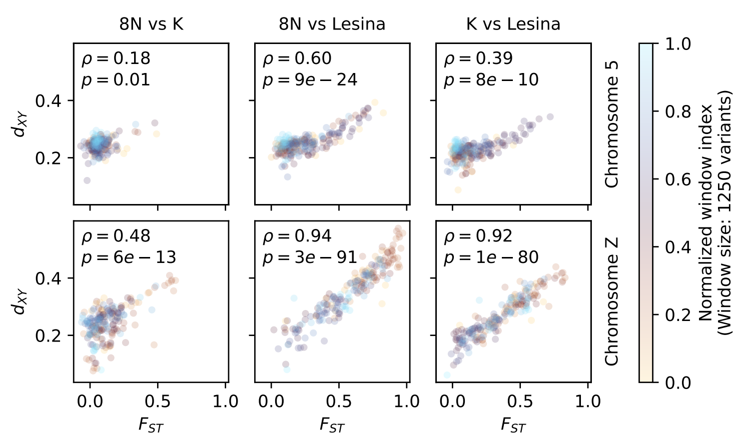

Figure 1: Spearman's ρ estimates were significant. Correlation was especially high in pairwise F_{ST} and d_{XY} between Lesina cohorts and others on the Z chromosome. Window indices were normalized by the total number of windows per chromosome.

The Lesina cohort's Z chromosome that have a high proportion of differing bases compared to the other cohorts also maintain that difference even when diversity within their cohort is taken into consideration, and the same agreement is observed in the lower F_{ST} and d_{XY} values. Taken as a whole, the data suggest that d_{XY} are correlated to F_{ST}, although they might not be proxies of one another in all cases.

References

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